

DVD Store 2.1 Documentation

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DS2.1 features added by Girish Khadke

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1. Introduction

The DVD Store Version 2.1 (DS2) is a complete open source online e-commerce test application, with a backend database component, a web application layer, and driver programs. The goal in designing the database component as well as the midtier application was to utilize many advanced database features (transactions, stored procedures, triggers, referential integrity) while keeping the database easy to install and understand. The DS2 workload may be used to test databases or as a stress tool for any purpose.

The distribution includes code for SQL Server, Oracle and MySQL. Included in the release are data generation programs, shell scripts to build data for any size DVD Store database as well as three standard sizes, database build scripts and stored procedure, PHP web pages, and a C# driver program.

Load data for the small DVD Store database is included in the kit.

The DS2 files are separated into database-independent data load files under ./ds2/data_files and driver programs under ./ds2/drivers and database-specific build scripts, loader programs, and driver programs in directories

./ds2/mysqls2
./ds2/oracleds2
./ds2/sqlserverds2

The DVD Store kit is available at linux.dell.com/dvdstore

file ds2.tar.gz contains ./ds2/data_files and ./ds2/drivers

file ds2_mysql.tar.gz contains ./ds2/mysqls2

file ds2_oracle.tar.gz contains ./ds2/oracleds2

file ds2_sqlserver.tar.gz contains ./ds2/sqlserverds2

To install:

In the directory in which you want to install ds2:

```
tar -xvzf ds2.tar.gz
```

and then install the implementation(s) of interest:

```
tar -xvzf ds2_mysql.tar.gz
```

```
tar -xvzf ds2_oracle.tar.gz
```

```
tar -xvzf ds2_sqlserver.tar.gz
```

WinZip may be used to unzip the programs for Windows - ignore the error message

The loader programs use relative addressing to reference the data files. They will need to be changed if the data files are placed elsewhere.

DS2 comes in 3 standard sizes:

Database	Size	Customers	Orders	Products
Small	10 MB	20,000	1,000/month	10,000
Medium	1 GB	2,000,000	100,000/month	100,000
Large	100 GB	200,000,000	10,000,000/month	1,000,000

Other sizes are easily created using the utilities described below

ds2.tar.gz contains data files for the Small version.

Most of the directories contain readme's with further instructions

2. What's New in 2.1

There are some great new changes and enhancements in the DVDStore 2.1. Further details about these changes are in this document:

Creation of any size DB - previous versions only supported S/M/L (10MB, 1GB, 100GB), but now any size can be specified. This feature is enabled through a Perl script that gathers the needed information and then outputs a set of scripts that will create the custom database including database build scripts and load files. The driver program was modified to be able to drive load against these custom size databases.

Automated creation of driver program parameter file - A Perl script can be run that asks the user questions and then outputs a parameter file that can be used to run the driver program against the database.

Driver Program can be run against multiple targets - A single instance of the driver program can run against multiple targets. It keeps the

stats for each target and also reports the totals throughout the run.

SQL Server parameterized query optimization - A SQL query in the purchase phase was converted to a parameterized form that results in more efficient execution. Increases OPM by approximately 20% while lowering CPU utilization when compared with the previous version of DVD Store.

CPU utilization for Linux target hosts can be gathered and reported by the driver program.

SpringSource web tier for Oracle DB added - In addition to the existing web tier implementations included with the DVDStore a SpringSource based version for the Oracle DB was created.

Updated dates for order data to current year.

3. Getting Started Quickly

Simple instructions for building the small DVDStore database using the data included in the kit without use of the Perl scripts are given in each database directory, for example, `.\ds2\sqlserverds2\ds2_sqlserver_readme.txt`

This section gives an overview of the steps needed to get a DVDStore of any size up and running quickly using the new Perl scripts.

Here is a short list of steps that can get you started quickly. Additional details are given later in this document.

- (a) Install database system (OS + Oracle or SQL Server or MySQL)
- (b) Download DVDStore2.1 to database system
- (c) Ensure that Perl is setup on system
- (d) Run `InstallDVDStore.pl`
- (e) Use scripts created by `InstallDVDStore.pl` to create and load database
- (f) On system to be used as driver, run `CreateConfigFile.pl`
- (g) Run driver program with the parameter file created by

`CreateConfigFile.pl`

4. Prerequisites

In order to run the DVDStore, one of the supported databases (SQL Server, Oracle, or MySQL) must already be installed and configured.

4.1 Prerequisites:

4.1.1 Running Perl scripts on Windows machine:

In order to be able to run the DS 2.1 Perl scripts on a Windows machine that generate data and build scripts for your database, you need to have Perl installed. The scripts have been tested with Perl from ActiveState and Cygwin

ActiveState:

(a) The free Community Edition may be downloaded at <http://www.activestate.com/activeperl/downloads>. Select ActivePerl for the version of Windows you have and run the downloaded file to install Perl

Cygwin:

- (a) Download setup.exe from <http://www.cygwin.com/> for cygwin installation.
- (b) Run the installer(setup.exe) to install cygwin.
- (c) Installer for cygwin will ask for mirrors from which to choose to copy setup files from.
(If not successful, retry by selecting another mirror since some of mirrors may not be active)
- (d) After selecting mirror, installer will ask for which packages to install.
(These packages are for emulating some of linux commands including Perl)
- (e) Select packages for gcc and Perl and their dependent packages.
(Or you can just select default package and select Perl checkbox also.
- Note: Perl and gcc are not included in default package and must be selected in addition to the defaults.)
- Note2: gcc is required to generate the best quality data on Windows
- (f) After the setup is successful, add the cygwin bin folder to the "Path" environment variable.
E.g. If cygwin is installed in folder c:\cygwin\ then set path variable as c:\cygwin\bin\
- (g) Make sure that above steps are successfully completed.
- (h) Now you can open cygwin shell and navigate to ds2 directory and run the Perl scripts.

Alternatively you can also run Perl scripts from DOS Shell successfully if you have installed cygwin and Perl.

4.1.2 Capturing CPU Utilization on Linux based target machines:

In previous versions of the DVD Store is was possible for the driver program to include the CPU utilization of the host in its output only if the target was a windows system. In DS2.1 it is now possible for users to get CPU utilization of remote linux machines.

If user is interested in getting CPU utilization for a Linux host from driver program the following is required :

- (a) Download plink.exe from <http://www.putty.org/> and place in a directory on Client Driver System that is set as the

environment variable PLINK_PATH.

For example: SET PLINK_PATH=C:\plink\plink.exe

Note: Download plink.exe and set environment variable on the machine from which driver program is run.)

(b) sysstat package needs to be installed on target linux machines. "sysstat" package is required because it includes mpstat command which is used to get CPU utilization by the driver program.

(c) Driver program parameter linux_perf_host parameter be set in following format:

<username>:<password>:<IP Address/hostname>

The username used should be a root user.

Note: you may have to run driver program twice to get Linux performance counters enabled

4.1.3 The Oracle driver program ds2oracledriver.exe is now compiled with the 64b Oracle 11g Oracle Data Provider for .NET

For backward compatibility the 32b client driver has been retained with the filename ds2oracledriver_32b_client.exe and the source file has been renamed ds2oraclefns_32b_client.cs.

5. Using Perl scripts for data generation, build script and configuration file generation

New Perl scripts that automate the task of generating CSV data files have been added with DVDStore2.1. They create database build and cleanup scripts and configuration files which will be used to run the driver program.

There are two Perl scripts : InstallDVDStore.pl and CreateConfigFile.pl

5.1 InstallDVDStore.pl

This Perl script will ask the following:

- (1) Database Size as an integer (User can enter any integer value. For example: 20, 40, 50, 100, 150, etc)
- (2) Database Size in MB/GB (Defines if the size entered initially was in mb or gb. Possible values are: mb, gb, MB, GB)
- (3) Database Type (User has to specify one of the following values: mssql (for SQL Server), mysql (for MySQL) or oracle (for Oracle))
- (4) System OS Type (User has to specify one of the following values: win (for windows) or linux(for linux))

(5) Database file paths (In case of Oracle and SQL Server database, user has to specify paths where the database files for primary dbfile, tables, indexes and log will be stored.

For multiple paths a ';' separated list can be provided.

For MySQL database, there is no need to enter this parameter)

For example the following provides 4 different paths:

c:\sql\;d:\sql\;e:\sql\;f:\sql\

This Perl script generates CSV text files to be used to load as data in the DVD Store database and are structured based on the DVD Store schema.

These files are stored in the cust, orders, and prod directories under the data_files directory.

This Perl script also generates customized build script files for database type selected under the database specific directory.

The database build scripts are created based on the following set of template files that are included in DVDStore 2.1:

(1) MySQL

[1] mysqls2_cleanup_generic_template.sql	in
/ds2/mysqls2/build folder	

(2) Oracle

[1] oracles2_create_all_generic_template.sh	in
/ds2/oracles2 folder	
[2] oracles2_cleanup_generic_fk_disabled_template.sh	in
/ds2/oracles2/build folder	
[3] oracles2_cleanup_generic_fk_disabled_template.sql	in
/ds2/oracles2/build folder	
[4] oracles2_cleanup_generic_template.sh	
in /ds2/oracles2/build folder	
[5] oracles2_cleanup_generic_template.sql	in
/ds2/oracles2/build folder	
[6] oracles2_create_db_generic_template.sql	in
/ds2/oracles2/build folder	
[7] oracles2_create_tablespace_generic_template.sql	in
/ds2/oracles2/build folder	

(3) SQL Server

[1] sqlserverds2_create_all_generic_template.sql	in
/ds2/sqlserverds2 folder	
[2] sqlserverds2_cleanup_generic_template.sql	in
/ds2/sqlserverds2/build folder	
[3] sqlserverds2_create_db_generic_template.sql	in
/ds2/sqlserverds2/build folder	
[4] sqlserverds2_create_ind_generic_template.sql	in
/ds2/sqlserverds2/build folder	

Build script files are generated from template files which have _generic_template in file name.

The build scripts generated from these template files will be have _databasesize in file name.

For example: For 20 mb sql server database instance, a file generated from template file sqlserverds2_create_all_generic_template.sql will be named: sqlserverds2_create_all_20mb.sql

New scripts are generated in the same folder in which their respective template files reside.

*Notes : (1) The primary reason that build scripts are created, instead of just directly running the commands, is that the user can always edit them later. It is very easy to modify these files to account for changing circumstances like changing paths where DBFiles(primary, index or table) or full text index files will be stored.

(2) User can also change the default size of DBFile size and default size of increments of growth in case of overflow in size.

(3) In case of SQL Server database case, user can specify 7 different database file paths for primary dbfile, misc dbfile, customer table, order table dbfile, index dbfile, log file and full text index file respectively. The Template script sqlserverds2_create_all_generic_template.sql assumes two dbfiles per table. In case if user wants to change number of dbfiles and file size or paths, user can always edit newly created script.)

If user specifies only one path, Perl script will assume same paths for all dbfiles mentioned above and will create database build script with same path name for all dbfiles.

(4) For Oracle, user can specify 4 paths for following tablespaces respectively: customer, index, misc and order.

If user specifies only one path, Perl script will assume same paths for all dbfiles mentioned above and will create database build script with same path name for all dbfiles.

(5) All paths should be trailed by following character : \ in case of windows and / in case of Linux.

For Example: For windows paths can be c:\sql\dbfiles\;d:\sql\dbfiles\

For Linux paths can be /sql/dbfiles/

(6) Multiple paths will always be separated by character ;

After execution of this Perl script, user needs to run CreateConfigFile.pl Perl script to generate configuration file used to drive

Driver program.

5.2 CreateConfigFile.pl

This Perl script asks for the following:

(1) Target host(s) (User can enter single or multiple targets to drive workload on. In case of multiple targets, ; character will separate each target)

(1a) When creating a config file for a web server target, the web server portnumber can be included in the "target" parameter.

The format to use for this is <hostname/ip>:<portnumber> where the portnumber is optional.

(2) Database size (User has to specify the database instance size created on target host specified. For example: 5gb, 20gb, etc. and all the targets be the same size. If different sizes are needed, then multiple driver programs must be run.)

(3) Target hostname for perfmon display for windows targets. This is an optional parameter.

(4) Target linux machine username,password and IP address or machine name (This parameter uses this format <username>:<password>:<IP Address> For multiple linux targets, semi colon ; separated list can also be given)

(5) User has to specify value for detailed_view parameter which is used to show or not to show detailed view of statistics for each target.

(User has to specify Y (for yes) and N (for No) as an answer)

This Perl script will create a configuration file named DriverConfig.txt under \ds2 folder. Once this file is created, user can edit this file for parameters like n_threads,run_time, think_time, etc.

6. Running driver programs using configuration file created

After all above steps and database creation, user can run driver program from command prompt as follows:

```
ds2sqlserverdriver.exe --config_file=c:\ds2\DriverConfig.txt
```

Driver program can still be run without specifying a config file. Instead user will have to provide all command line parameters to driver program as parameter.

7. Driving workload on multiple target machines:

In DVD Store v2.0, driver programs were able to only drive workload against a single machine with a Database Server or Web Server installed.

In v2.0, in order to drive workloads on multiple systems with Database or Web Servers installed, multiple instances of driver programs had to be opened and run at same time to drive workload against multiple machines

or targets.

In DVD Store v2.1, we have modified the Driver programs to be able to distribute the threads equally against multiple target machines specified either on the command line or in a configuration file.

In addition to showing the total stats across all targets, the statistics for each individual machine are collected by the driver program and will be displayed on the console based on the "detailed_view" parameter specified in the configuration file. If "detailed_view" parameter is set to value "Y", driver program will print individual as well as aggregate statistics (like opm, response times, CPU%, etc) for each target machine on every 10 seconds interval as well as after the run completes.

If value of parameter "detailed_view" is "N", driver program will print aggregate or total statistics after every 10 second interval and print total aggregate as well as individual statistics for each target machine after run completes.

The prerequisites for driving workload on multiple machines are as follows:

- (1) Each machine has database instance of same size.
- (2) Threads will get spread equally on each machine. e.g. If "n_threads" parameter is set as 2 in configuration file and configuration file has two target machine names, overall 4 threads will be spawned and will spread equally among two machines which is 2 threads per machine.

Driver program can be run as follows:

```
ds2webdriver.exe --config_file=c:\ds2\DriverConfig.txt
```

Contents of Config file DriverConfig.txt will look like as follows:

```
target=10.115.66.150;10.115.66.134
n_threads=4
ramp_rate=10
run_time=1
db_size=5gb
warmup_time=1
think_time=0.85
pct_newcustomers=20
n_searches=3
search_batch_size=5
n_line_items=5
virt_dir=ds2
page_type=aspx
windows_perf_host=10.115.66.150;10.115.66.134
detailed_view=Y
```

Above statement will case web driver to drive workload against two target windows machines with 5gb database instances and spawn 2 threads per machine and print individual as well as aggregate statistics on console.

For getting linux CPU Utilization data, linux_perf_host parameter must be specified.

Sample Configfile to run driver against multiple linux targets will look like as follows:

```
target=10.115.66.150;10.115.66.134
n_threads=4
ramp_rate=10
run_time=1
db_size=5gb
warmup_time=1
think_time=0.85
pct_newcustomers=20
n_searches=3
search_batch_size=5
n_line_items=5
virt_dir=ds2
page_type=php
linux_perf_host=root;qazwsx:10.115.66.150;root:xswzaq:10.115.66.134
detailed_view=Y
```

8. Modifications to Driver Programs:

8.1 Using custom database size:

In DVD Store v2.0, driver program was able to drive workload on only standard database sizes S | M | L (10mb | 1gb | 100GB).

However in current version of DVD Store, we have modified driver program such that it can drive workload on custom database sizes.

The database size can be specified by a parameter through configuration file : "db_size". Values of parameter "db_size" can be :

S | M | L (to maintain backward compatibility with standard size databases) or any value of database size like 200mb, 20gb, 150gb, etc.

In case of driving database workloads on multiple target machines, each machine should have database instance of same size.

8.2 Parameterization of query for SQL Server driver:

In the case of SQL Server it was discovered that a single query that is created during the purchase phase of the order was a large

percentage of the plan cache. This was because although the query was essentially the same each time, SQL Server did not recognize

it as the same and was instead creating a new plan and storing it in the plan cache each time. By converting the way the query was

created so that it was a parameterized query SQL Server then only created a plan once and then reused it for every order.

This change is significant because it changes the workload for SQL Server

from the previous version of the DVD Store. This change was found to increase the OPM by as much as 20%.

8.3 Additional Error Handling

The web driver program was also modified in a couple of places to more gracefully handle unexpected http errors. A max timeout of 60 seconds was added for initial connection of all threads. If all threads are not connected in 60 seconds, the driver program will exit.

9. Modifications to Data generation C programs

In DVD Store v2.0, data generation C programs generate data for standard size instances (S | M | L) only. After generating CSV files, they needed to be converted into proper CR/LF format if they were to be used on a windows system with unix2dos conversion tool.

In current version of DVD Store, we have modified C programs to generate CSV files for any database size. Also there is no need to convert generated CSV files which saves a lot of time for converting huge CSV files into required format. If you use the CreateConfig.pl script the data generation will be run automatically and it will not be necessary to run these commands. The information is provided here as additional detail for those interested.

Following programs were modified:

(1) ds2_create_cust.c

New command line argument is introduced for this c program : n_Sys_Type which denoted system type for which CSV files needs to be generated in correct format. A 0 for linux and 1 for Windows.

This C program can be run as follows:

```
./ds2_create_cust      1  10000 US   S 0 > us_cust.csv &  
./ds2_create_cust 10001  20000 ROW  S 0 > row_cust.csv &
```

Above statements will generate customer data file for 10mb instance for linux machine.

(2) ds2_create_inv.c

New command line argument is introduced for this c program : n_Sys_Type which denoted system type for which CSV files needs to be generated in correct format.

(0 for linux and 1 for Windows).

This C program can be run as follows:

```
./ds2_create_inv 10000 0 > ../prod/inv.csv
```

Above statement will generate inventory data file for 10mb instance for linux machine

(3) ds2_create_orders.c

Following new command line arguments are introduced for this C program:

n_Sys_Type (0 for linux and 1 for windows)
n_Max_Prod_Id (max product Id for custom database size)
n_Max_Cust_Id (max customer Id for custom database size)

This C program can be run as follows:

```
./ds2_create_orders 1 1000 jan S 1 0 10000 20000 &  
./ds2_create_orders 1001 2000 feb S 2 0 10000 20000 &  
./ds2_create_orders 2001 3000 mar S 3 0 10000 20000 &  
./ds2_create_orders 3001 4000 apr S 4 0 10000 20000 &  
./ds2_create_orders 4001 5000 may S 5 0 10000 20000 &  
./ds2_create_orders 5001 6000 jun S 6 0 10000 20000 &  
./ds2_create_orders 6001 7000 jul S 7 0 10000 20000 &  
./ds2_create_orders 7001 8000 aug S 8 0 10000 20000 &  
./ds2_create_orders 8001 9000 sep S 9 0 10000 20000 &  
./ds2_create_orders 9001 10000 oct S 10 0 10000 20000 &  
./ds2_create_orders 10001 11000 nov S 11 0 10000 20000 &  
./ds2_create_orders 11001 12000 dec S 12 0 10000 20000 &
```

Above statements will generate orders data files for 10mb instance for linux machine.

(4) ds2_create_prod.c

Following command line argument is introduced for this c program:

n_Sys_Type (0 for linux and 1 for windows)

This C program can be run as follows:

```
./ds2_create_prod 10000 0 > prod.csv
```

Above statement will generate products data file for 10mb instance for linux machine.

NOTE: It is also possible to copy the data files back and forth from Windows to Linux using the text or ascii filetype in programs such as WINSCP

10. Running SQL Server Maintenance Task on SQL Server 2008

Once the database build and bulk load finishes for SQL Server 2008, in order to get better performance, it is necessary to update

statistics of tables in databases DS2. In previous versions of SQL Server (SQL Server 2005 and 2000), sqlmaint.exe, a command line utility that came with database server installation, was used. This utility is now obsolete and is no longer included with SQL Server 2008.

A maintenance task needs to be created to accomplish the same function in SQL Server 2008.

In SQL Server 2008 Management Studio (GUI), follow following steps:

- 1) Go to Object Explorer and click and expand database server tree.
- 2) Under server tree, expand management and right click on maintenance plans.
- 3) Left Click on "Maintenance Plan Wizard Option".
- 4) In the wizard opened, click next and enter name of plan as "ds2".
- 5) Click next and check "Update Statistics" checkbox and again click next.
- 6) Click next and then choose database as DS2 and click OK.
- 7) Ensure "All existing statistics" and "Sample By" checkbox are set along with value "18" "percent".
- 8) Once above step is done click next twice to create a task under "Maintenance Plans" under "Management" object under SQL Server tree.
- 9) Now right click on this task "ds2" created from above steps and it will show a menu option for right click.
- 10) Click execute to update statistics on all tables in DS2 database using task created due to above steps.

For previous versions of SQL Server, user can directly invoke sqlmaint.exe utility from command prompt as specified in old documents.

11. Using SpringSource Web Tier for Oracle database testing using web driver

In DVDStore 2.1, SpringSource MVC Implementation of Web Pages for Oracle database is provided along with old JSP implementation.

If user wants to use SpringSource web tier instead of JSP, here are the steps to ensure Web Tier is setup correctly on Web Server:

- 1) To compile SpringSource code, we need JDK to be installed on Web Server preferably JDK 1.6 or later.
- 2) After installing set environment variable JAVA_HOME to path where JDK is installed.
e.g. JAVA_HOME will have value C:\Program Files\Java\jdk1.6.0_20\
- 3) Also Install Ant to compile SpringSource code using ant build commands.
- 4) After installing ant, set ANT_HOME environment variable to path where ant is installed.
e.g. ANT_HOME will have value C:\apache-ant-1.8.1\
- 5) Add similar entries for PATH environment variable too.
e.g. For java, the entry in PATH environment variable will be C:\Program Files\Java\jdk1.6.0_20\bin\
For ant, the entry in PATH environment variable will be

C:\apache-ant-1.8.1\bin\

6) Install Apache Tomcat Web Server (Version 5.5 or above)

7) Set CATALINA_HOME environment variable to path where Tomcat is installed.

e.g. CATALINA_HOME will have value C:\Program Files\Apache Software Foundation\Tomcat 5.5\

8) Download OpenSource SpringSource Framework 2.5.5 from following link:

<http://sourceforge.net/projects/springframework-2/2.5.5/>

(Download Framework with dependencies)

9) Unzip the zip file downloaded in above step in a folder on system on which web server is installed.

e.g. Unzip above zip file into folder

c:\springsource-framework2.5.5\

10) Go to /ds2/oracleds2/web/springsource/ and copy ds2web folder into webapps directory of Tomcat 5.5 installation directory.

e.g. Copy ds2web folder in DVDStore 2.1 from

C:\ds2\oracleds2\web\springsource\ to

C:\Program Files\Apache Software Foundation\Tomcat 5.5\webapps\ directory.

11) Copy following files from SpringSource Framework to ds2web folder in webapps directory of Tomcat i.e. /ds2web/WEB-INF/lib/.

FileName

CopyFrom Path

CopyTo Path

spring.jar

C:\springsource-framework2.5.5\dist\spring.jar

C:\Program Files\Apache Software Foundation\Tomcat 5.5\webapps\ds2web\WEB-INF\lib\spring-webmvc.jar

C:\springsource-framework2.5.5\dist\modules\spring-webmvc.jar C:\Program Files\Apache Software Foundation\Tomcat 5.5\webapps\ds2web\WEB-INF\lib\standard.jar

C:\springsource-framework2.5.5\lib\jakarta-taglibs\standard.jar C:\Program Files\Apache Software Foundation\Tomcat 5.5\webapps\ds2web\WEB-INF\lib\commons-logging.jar

C:\springsource-framework2.5.5\lib\jakarta-commons\commons-logging.jar C:\Program Files\Apache Software Foundation\Tomcat 5.5\webapps\ds2web\WEB-INF\lib\servlet-api.jar

C:\springsource-framework2.5.5\lib\j2ee\servlet-api.jar C:\Program Files\Apache Software Foundation\Tomcat 5.5\webapps\ds2web\WEB-INF\lib\jstl.jar

C:\springsource-framework2.5.5\lib\j2ee\jstl.jar

C:\Program Files\Apache Software Foundation\Tomcat 5.5\webapps\ds2web\WEB-INF\lib\

Also Obtain Oracle J Connector (ojdbc14.jar) from

http://www.oracle.com/technology/software/tech/java/sqlj_jdbc/index.html

Copy ojdbc14.jar to lib folder under ds2web folder in Apache Tomcat webapps. i.e. C:\Program Files\Apache Software Foundation\Tomcat 5.5\webapps\ds2web\WEB-INF\lib\

12) Open command prompt and go to /webapps/ds2web/WEB-INF/ folder under tomcat installation.

13) Type "ant clean" and press enter. (This step will clean up the older *.class files under ds2web folder)

14) Type "ant" and press enter. (This step will build all code under ds2web folder using build.xml file in /ds2web/WEB-INF/ and will dump *.class files in their respective folders)

15) For Web Driver to work for Oracle Web Tier, it is necessary to create two triggers on Database Server.

The script to create triggers is located at
/ds2/oracleds2/web/springsource/oracleds2_create_trigger_springsource_only.sql

Run this script on command prompt as: sqlplus ds2/ds2
@oracleds2_create_trigger_springsource_only.sql

16) One build using ant is complete and all above steps are complete, start apache tomcat web server and check whether DVDStore Web tier is up and running using following URL:

http://<hostname>:8080/dslogin.html

17) Now to run the web driver program and with SpringSource Web pages being hosted by Tomcat, some the parameter values for driver program should be as follows:

virt_dir=ds2web
page_type=html